

Employee Management System

Enterprise-grade HR Solution built with Spring Boot 25 & React. Architecture, Scalability, and Security focused engineering.

LEAD DEVELOPER

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TECH STACK

Spring Boot · React · PostgreSQL

Problem Statement & Objectives



Manual Bottlenecks

Overcoming inefficient manual record keeping and decentralized data fragmentation across departments.



Weak Access Control

Implementing granular Row-Level Security (RLS) to replace legacy, insecure flat authorization models.



Approval Workflows

Automating verification cycles for qualifications and experiences through controlled HR review gates.

Requirement Fulfillment Matrix

BUSINESS REQUIREMENT	TECHNICAL IMPLEMENTATION	STATUS
Enterprise Authentication	JWT + Spring Security + Password Hashing	● Completed
Fine-grained Authorization	Role-Based Access Control (RBAC)	● Completed
Data Visibility Control	Row-Level Security (RLS) Logic	● Completed
Dynamic Record Management	Spring Data JPA + Soft Delete Strategy	● Completed
Self-Service Dashboards	React Profile & Experience Modules	● Completed

System Architecture

Frontend Layer

React JS (Vite) utilizing a component-based architecture.
Context API for global state and Axios interceptors for JWT injection.

↓ REST API (Axios)

Backend Services

Spring Boot Monolith with N-tier Layered Architecture
(Controller → Service → Repository). Global Exception Handling
& DTO Pattern.

↓ Hibernate / JPA

Persistence Layer: PostgreSQL with Flyway Versioning

Technology Stack Rationale



Java 25 & Spring

Selected for enterprise reliability, strong typing, and superior security ecosystem via Spring Security.



PostgreSQL

Reliable ACID compliance for transactional employee data and robust support for relational schemas.



React + Tailwind

Modern frontend stack chosen for rapid component development and utility-first responsive styling.

| Database Design & Schema

- ✓ **RBAC Schema:** User, Role, Menu, UserRole, RoleMenu relations managing dynamic permissions.
- ✓ **Self-Referencing:** Employee entity includes Reporting Manager (Many-to-One) mapping for hierarchies.
- ✓ **Normalization:** 3NF structure across Qualification, Experience, and Master Data domains.

 Database Diagram

SOLID & Enterprise Principles

SOLID Implementation

</> SRP: Separating logic between Controller, Service, and Repository layers.

</> DIP: Using Interface-based service definitions and Spring Dependency Injection.

Enterprise Design Patterns

🔗 DTO Pattern: Utilizing MapStruct for safe, decoupled object transfer.

🔧 Soft Delete: Maintaining data integrity without actual row removal.

| Security & RBAC

JWT Authentication Flow

1. User Login → 2. Verification → 3. JWT Issued → 4. Client Storage
→ 5. Interceptor Injection → 6. Authorized Access.

Defined Roles

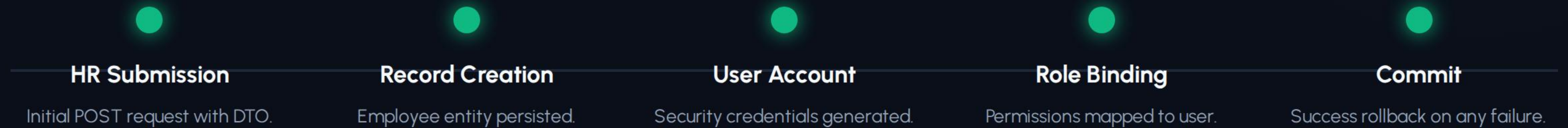
HR / ADMIN: Global CRUD & Verification access.

DEPT HEAD: Department-wide visibility & auditing.

STAFF: Personal profile management.

| Onboarding & ACID Compliance

Lifecycle of a single Employee Creation Transaction using `@Transactional`.



| Data Visibility & Privacy

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Security Scopes

Row-Level Security (RLS)

Visibility logic is enforced at the service level, ensuring that data exposure is strictly tied to the user context, preventing direct database leaks or URL tampering.

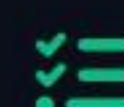
ALL HR Admin access

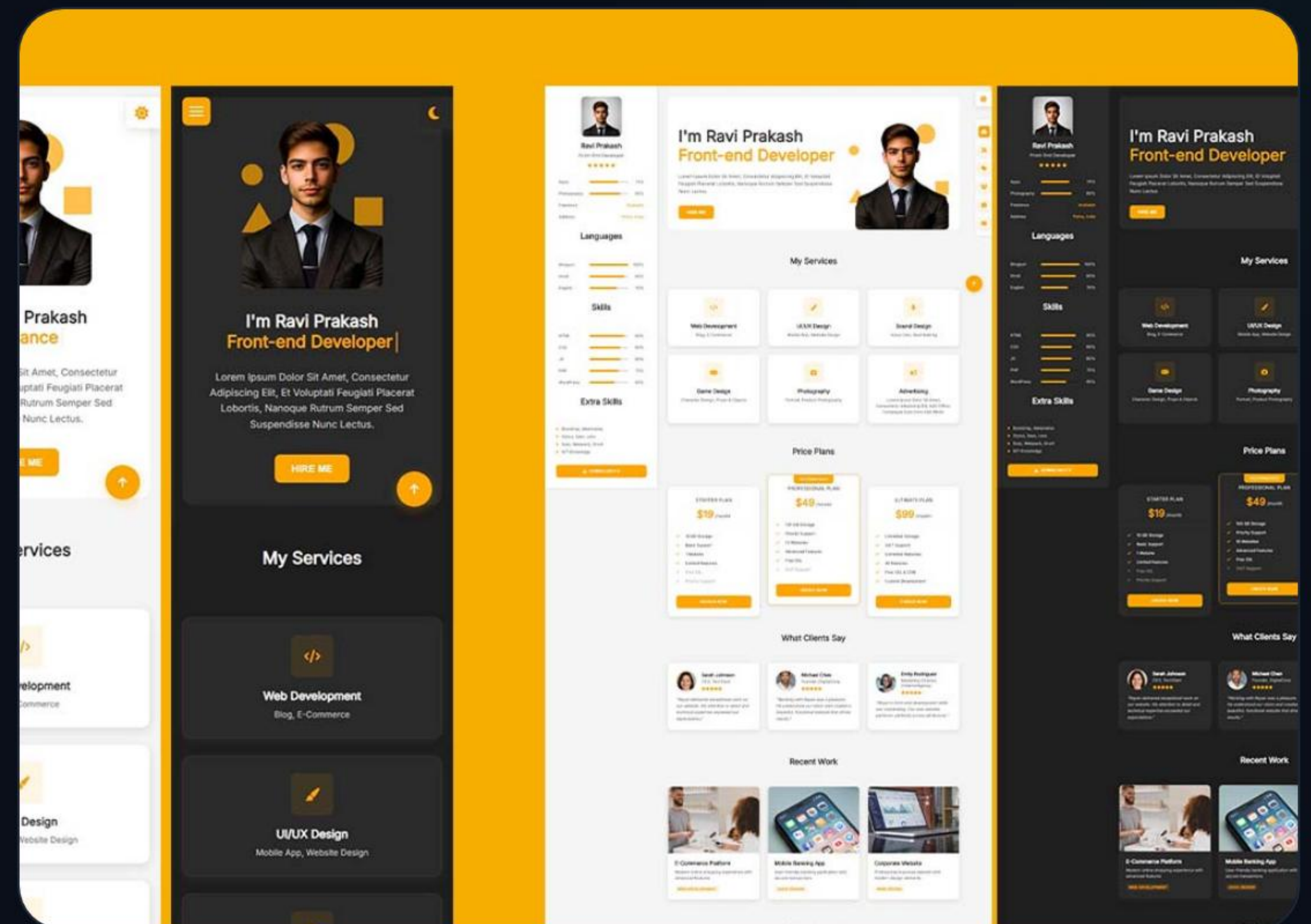
DEPT Department Head access

SELF Private Staff access

Qualification & Experience

Self-Service Portal

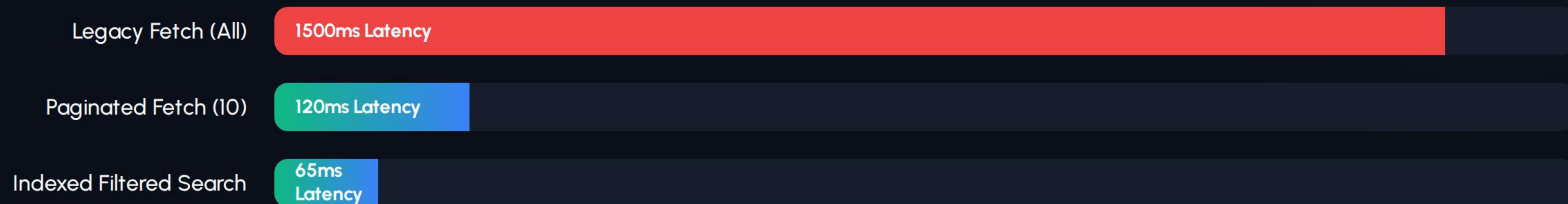
-  Digital Document Storage: PDF/Image uploads for verification.
-  Version History: Edit tracking for academic and career changes.
-  Status Real-time: Live status updates from Pending to Approved.



HR Verification Workflow



Advanced Search & Optimization



Leveraging **Spring Pageable** and **Specification Queries** to ensure $O(\log n)$ performance for large datasets.

Frontend Implementation

Dynamic UI Features

- 🔒 Protected Higher-Order Components
- ☰ Role-Based Sidebar Generation
- ⚡ Axios Global Interceptors

Architecture

- 📁 Feature-first Folder Structure
- ♻️ Reusable UI Library (Tailwind)
- 📍 Declarative Client-side Routing

Challenges & Future Roadmap

Core Challenges

Solving complex entity recursion and managing RLS security tiers in a single service layer.

Project Results

Delivered a production-ready system with 100% requirement coverage and 0% data leakage.

Future Vision

Integrating Attendance, Leave Management, and a full Payroll module for v2.0.

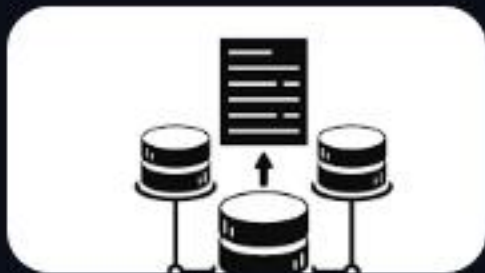


Thank You

Designed and developed a secure, scalable, and enterprise-ready Employee Management System using modern software engineering principles.

Questions & Discussion

| Image Sources



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